

Introduction to Quantum Spin Models

- Spin lattice models such as the Ising and Heisenberg models are fundamental in condensed matter physics, describing magnetism and correlated quantum many-body systems

[citation : 0en.wikipedia.org](https :
//en.wikipedia.org/wiki/Transverse-field_Ising_model : : text =
Image These models have non-commuting terms (e.g. $Z_i Z_j$ vs X_i) that
make classical simulation intractable for large lattices

[citation : 2en.wikipedia.org](https :
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Image

- Quantum computers can efficiently encode the 2^N -dimensional Hilbert space of N spins; variational algorithms like VQE are promising for finding ground states on near-term (NISQ) devices

[citation : 3learning.quantum.ibm.com](https :
//learning.quantum.ibm.com/tutorial/variational-quantum-
eigensolver : : text = Variational

2D Transverse-Field Ising Model

- The 2D transverse-field Ising Hamiltonian on a square lattice is given by

$$H_{\text{Ising}} = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_i X_i,$$

where $\langle i,j \rangle$ denotes nearest neighbors, J is the coupling, and h the transverse field strength [citation : 4en.wikipedia.org](https://en.wikipedia.org/wiki/Transverse-field_Ising_model : : text = ImageThe first term ($Z_i Z_j$) favors aligned or anti-aligned spins on adjacent sites; the second term (X_i) applies a perpendicular magnetic field.

- Here X_i, Z_i are Pauli matrices on qubit (spin) i . The competition of these non-commuting terms leads to a quantum phase transition in large lattices.

2D Isotropic Heisenberg Model

- The 2D isotropic Heisenberg model on a square lattice is

$$H_{\text{Heis}} = J \sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j + Z_i Z_j),$$

with the same coupling J for XX , YY , and ZZ interactions.

- This $SU(2)$ -symmetric model describes exchange interactions between neighboring spins. It is a prototypical model for quantum magnetism.
- On a 2×2 lattice (4 spins), each spin has 2 neighbors (in an open boundary scenario, there are 4 edges in total).

Variational Quantum Eigensolver (VQE)

- VQE is a hybrid quantum-classical algorithm for finding the ground state energy of a Hamiltonian

[citation : 5learning.quantum.ibm.com](https :

//learning.quantum.ibm.com/tutorial/variational – quantum – eigensolver : : text =

Variational A parameterized quantum circuit $U(\theta)$ (the ansatz) prepares a trial state $U(\theta)|0\rangle$; the cost function is the energy expectation

$$E(\theta) = \langle 0 | U^\dagger(\theta) H U(\theta) | 0 \rangle,$$

which is iteratively minimized [citation : 6nature.com](https :

//www.nature.com/articles/s41467 – 024 – 46402 – 9?error = cookies_not_supportedcode =

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- Classical optimizers update the parameters θ to drive $E(\theta)$ toward the ground-state energy. VQE requires fewer coherent gates than quantum phase estimation, making it suitable for NISQ devices

[citation : 7learning.quantum.ibm.com](https :

- In Qiskit, Hamiltonians are expressed as sums of Pauli operators using tensor products [citation : 9medium.com](https://medium.com/qiskit/introducing-qiskit-algorithms-with-qiskit-runtime-primitives-d89703ecfca3 : : text = hamiltonian One can construct a SparsePauliOp (or PauliSumOp) from a list of

- from

```
qiskit.quantum_info import SparsePauliOp
DefinePauliTerms for H1 singon 4 qubit
[("ZZII", -J), ("IZZI", -J), ("ZIZI", -J), ("IIZZ", -J), ("XIII", -h), ("IXII", -h)]
SparsePauliOp.from_list(paulis)
```

This leverages Qiskit's operator algebra to build H directly from Pauli products [citation : 10medium.com](https://medium.com/qiskit/introducing-qiskit-algorithms-with-qiskit-runtime-primitives-d89703ecfca3 : : text = hamiltonian

“begin-frame”–Ansatz Design and Optimization”

“begin-itemize”

“item We use the “texttt–TwoLocal” ansatz from Qiskit: alternating singlequbit rotation layers and entangling gates

[oai citation:11†medium.com](https://medium.com/qiskit/introducingqiskital

“item For instance use rotations around Y and Z axes (“texttt–ry” “texttt–rz”) on each qubit and controlledZ (“texttt–cz”) entangling gates. Layers (reps) can be increased for expressiveness.

“item In a 2×2 lattice entangling gates are applied along the edges (e.g. CZ between qubits 0–1 0–2 1–3 2–3) to reflect the lattice geometry.

“item The VQE is executed on a statevector simulator (e.g. “texttt–BasicAer.get`backend(‘statevector_simulator’))”withaclassicaloptimiz
“end-itemize”

“end-frame”

“begin-frame”[fragile]–Implementation: 2×2 Transverse Ising”

“begin-itemize”

“item Construct the 4qubit Hamiltonian and run VQE as follows:

“begin-verbatim”

from qiskit.quantum`info import SparsePauliOp

```

from qiskit.circuit.library import TwoLocal
from qiskit.algorithms import VQE
from qiskit.algorithms.optimizers import COBYLA
from qiskit import BasicAer
# Define Pauli terms for  $H = J \sum_i Z_i Z_{i+1} + h \sum_i X_i$ 
paulis = [
    ("ZZII" J) ("IZZI" J) ("ZIZI" J) ("IIZZ" J)
    ("XIII" h) ("IXII" h) ("IIXI" h) ("IIIX" h)
]
H = SparsePauliOp.from_list(paulis)
ansatz = TwoLocal(num_qubits=4 rotation_blocks=['ry''rz']
entanglement_blocks='cz' reps = 2)
result = VQE(ansatz COBYLA())
quantum_instance=BasicAer.get_backend('statevector_simulator')
).compute_minimum_eigenvalue(operator=H)
E0 = result.eigenvalue

```

This yields the estimated ground-state energy E_0 of the 2×2 Ising model. The exact ground energy (from diagonalizing H) can be used to verify accuracy.

Implementation: 2×2 Heisenberg

- Construct the Hamiltonian with XX , YY , and ZZ terms for each neighbor:

```
edges = [(0,1),(0,2),(1,3),(2,3)]
paulis = []
for (i,j) in edges:
    pX = ["I"]*4; pX[i] = "X"; pX[j] = "X"
    paulis.append(("".join(pX), J))
    pY = ["I"]*4; pY[i] = "Y"; pY[j] = "Y"
    paulis.append(("".join(pY), J))
    pZ = ["I"]*4; pZ[i] = "Z"; pZ[j] = "Z"
    paulis.append(("".join(pZ), J))
H = SparsePauliOp.from_list(paulis)

ansatz = TwoLocal(num_qubits=4, rotation_blocks=['ry','rz'],
                  entanglement_blocks='cz', reps=2)

result = VQE(ansatz, COBYLA(),
```


Ground State Energy vs Coupling

- By varying the coupling parameter (h for the Ising model, J for the Heisenberg model) and rerunning VQE (or exact diagonalization), one obtains the ground-state energy E_0 as a function of that parameter.
- For example, for a fixed $J = 1$ in the Ising model, compute $E_0(h)$ for h in some range (e.g. 0 to 2). Similarly, compute $E_0(J)$ for the Heisenberg model.
- These data points can be plotted to show the variation of E_0 with the coupling (see figure caption below). The resulting curves (monotonic in these simple cases) serve as a benchmark of the model's physics and the VQE's accuracy.
- Such plots are useful to observe trends or phase transitions in the ground state as parameters change, and to compare VQE vs exact results.

Lattice Connectivity and Circuit Illustration

- The 2×2 lattice connectivity forms a square: qubits 0–1–3–2 (with 0–2 and 1–3 as vertical edges, 0–1 and 2–3 as horizontal edges).
- In the quantum circuit, each qubit corresponds to a spin. Entangling gates (e.g. CZ or CNOT) are placed between qubits that are neighbors on the lattice to capture interactions.
- For instance, in the ansatz we apply CZ gates on pairs (0,1), (0,2), (1,3), (2,3) to reflect the 2D nearest-neighbor edges.
- (A circuit diagram would show single-qubit rotations on each qubit and CZ gates along the edges of a 2×2 grid, with the pattern of entanglement matching the lattice graph.)

Scaling and Limitations

- The Hilbert space dimension grows as 2^N for N spins, so exact simulation is intractable for large N . VQE helps but still faces challenges as system size grows.
- The number of variational parameters and required measurements increases with lattice size. Large parameter counts can lead to optimization difficulties (such as barren plateaus where gradients vanish) [citation: 12nature.com](https://www.nature.com/articles/s41467-024-46402-9?error=cookies_not_supported&code=e0d86c6c-d400-4a7e-9dda-2e6669633ed5 : text = Another NISQ limitations (finite coherence time, gate errors) restrict circuit depth and are prone.
- To date, most VQE demonstrations on spin models have been limited to small lattices (few qubits) [citation: 13nature.com](https://www.nature.com/articles/s41467-024-46402-9?error=cookies_not_supported&code=e0d86c6c-d400-4a7e-9dda-2e6669633ed5 : text = intrinsic Variational Quantum Simulation of 2D Spin Models)

References I

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